Mohd Qasim

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Academic Qualifications

Indian Institute of Technology, Jodhpur — Sep 2023 – May 2027
Bachelor of Technology in Chemical Engineering — *Jodhpur, Rajasthan*

Jawahar Navodaya Vidyalaya (JNV), Budgam — 2022
Class XII, CBSE — *Budgam, Jammu & Kashmir*

Mutahhary Public High School, Saliskote — 2020
Class X, JKBOSE — *Kargil, Ladakh*

Key Projects

- **DSA Visualizer: Interactive Algorithm Learning Platform** **[Live Site]**
  Team Project, Web Development.
  - Co-developed a web platform that animates **15+ data structures and algorithms** step by step, including 5 sorting algorithms, 7 graph algorithms (BFS, DFS, Dijkstra, Bellman–Ford, Prim, Kruskal, Topological Sort), and BST, AVL, B-Tree, Heap, Stack and Linked List.
  - Built canvas-based animations with **Konva.js** and modular ES6 JavaScript on a **Ruby on Rails** backend, with each structure written as a self-contained module so the team could add new visualizers in parallel.
  - Implemented playback controls (play/pause, forward and backward stepping, adjustable speed) with live narration of every comparison and swap, plus time and space complexity panels for each algorithm.
  - Visualized AVL rotations (LL, RR, LR, RL) and BST traversals (pre-, in-, post- and level-order), with zoom and pan support for exploring large trees.
  - Collaborated through Git and GitHub using feature branches and merges to refactor the codebase into reusable modules, and containerized the app with **Docker** for deployment.
- **Named Entity Recognition (NER)**
  Mentor: Prof. Avinash Sharma , Department of Computer Science Engineering.
  - Built an end-to-end Named Entity Recognition (NER) system using Python and scikit-learn, focusing on entity extraction from unstructured text.
  - Conducted thorough data preprocessing, including handling noisy characters, missing values, and sentence segmentation from raw token-level data.
  - Engineered custom features (e.g., token patterns, casing, word length) and trained a Random Forest classifier for sequence tagging.
  - Evaluated model with NER-specific metrics (precision, recall, F1-score) using the nervaluate library; plotted evaluation results by entity type.
  - Utilized seaborn, matplotlib, and pandas to present results and trends, enhancing model explainability.

Technical Skills

- **Programming Languages:** C, C++, Python, JavaScript
- **Web Development:** JavaScript (ES6), HTML5, CSS3, Konva.js, Ruby on Rails
- **Software and Libraries:** Git, GitHub, OOPS, Flask, Docker, Aspen Plus
- **Machine Learning Libraries:** Scikit-Learn, PyTorch, TensorFlow, Keras

Relevant Coursework

- **Computer Science:** Introduction to Computer Science, Introduction to Machine Learning
- **Mathematics:** Multivariable and Vector Calculus, Linear Algebra and Differential Equations, Probability, Statistics and Stochastic Processes
- **Core Engineering and Sciences:** Fluid Mechanics, Thermodynamics, Mass Transfer, Material and Energy Balance, Chemical Reaction Engineering, Molecular Engineering, Quantum Chemistry and Spectroscopy